

Compounds of Transitional elements

41. The equivalent weight of $K_2Cr_2O_7$ in acidic medium
 (a) 294 (b) 298
 (c) 49 (d) 50
42. The transitional element of III-B group is isomorphous with $Al_2(SO_4)_3$. Therefore the compound will be
 (a) Purple (b) Blue
 (c) White (d) Uncertain
43. A copper salt is isomorphous with $ZnSO_4$, the salt will be
 (a) Paramagnetic
 (b) Diamagnetic
 (c) Ferromagnetic
 (d) None
44. V_2O_5 is useful as catalyst in
 (a) Manufacture of H_2SO_4
 (b) Manufacture of HNO_3
 (c) Manufacture of Na_2CO_3
 (d) It is not a catalyst
45. $KMnO_4$ in basic medium is reduced to
 (a) K_2MnO_4 (b) MnO_2
 (c) $Mn(OH)_2$ (d) Mn^{2+}
46. When $KMnO_4$ is reduced with oxalic acid in acidic solution, the oxidation number of Mn changes from
 (a) 7 to 4 (b) 6 to 4
 (c) 7 to 2 (d) 4 to 2
47. Nessler's reagent is
 (a) K_2HgI_4
 (b) $K_2HgI_4 + KOH$
 (c) $K_2HgI_2 + KOH$
 (d) $K_2HgI_4 + Hg$
48. When ammonium dichromate is heated, the gas formed is
 (a) N_2 (b) O_2
 (c) H_2 (d) NH_3
49. Acidified potassium dichromate on reacting with a sulphite is reduced to
 (a) CrO_2Cl_2 (b) CrO_4^{2-}
 (c) Cr^{3+} (d) Cr^{2+}
50. The product of oxidation of I^- ion by MnO_4^- in alkaline medium is
 (a) I_2 (b) IO_3^-
 (c) IO_4^- (d) I_3^-
51. Identify the statement which is not correct regarding copper sulphate
 (a) It reacts with KI to give iodine
 (b) It reacts with KCl to give Cu_2Cl_2
 (c) It reacts with $NaOH$ and glucose to give Cu_2O
 (d) It gives CuO on strong heating in air
52. Acidified potassium permanganate solution is decolourised by
 (a) Bleaching powder
 (b) White vitriol
 (c) Mohr's salt
 (d) Microcosmic salt
 (e) Laughing gas
53. Which of the following oxides is white but becomes yellow on heating



- (a) AgO (b) Ag_2O (c) FeO (d) ZnO $Co = 27, Cu = 29, Ni = 28$). The colourless species are
54. Amalgams are
(a) Highly coloured alloys
(b) Always solid
(c) Alloys which contain mercury as one of the contents
(d) Alloys which have great resistance to abrasion
55. In photography sodium thiosulphate is used as
(a) Complexing agent
(b) Oxidising agent
(c) Reducing agent
(d) None of these
56. The substance that sublimes on heating
(a) Magnesium chloride
(b) Silver chloride
(c) Mercurous chloride
(d) Sodium chloride
57. $K_3[Fe(CN)_6]$ is called
(a) Potassium ferricyanide
(b) Red prussiate of potash
(c) Potassium hexacyanoferrate (III)
(d) All of these
58. Which of the following will show increase in weight when kept in magnetic field
(a) TiO_2 (b) $Fe_2(SO_4)_3$
(c) $KMnO_4$ (d) $ScCl_3$
59. Amongst TiF_6^{2-} , CoF_6^{3-} , Cu_2Cl_2 and $NiCl_4^{2-}$ (Atomic number $Ti = 22$,

